Internship Opportunity – 2D Localisation of Vibration Events Using DAS: Railway Applications

Background

Distributed Acoustic Sensing (DAS) turns an optical fibre into a continuous array of acoustic sensors capable of measuring vibrations over several kilometres. Its operation relies on Rayleigh backscattering: when a light pulse travels through the fibre, a small fraction of the light is scattered back towards the source due to microscopic irregularities in the glass. The phase and amplitude of the backscattered light change when the fibre is strained, making it possible to detect and locate vibrational events.
The SURFO team uses DAS to monitor railway infrastructure in real time. Applications include traffic monitoring, detection of track and rolling-stock defects, identification of external events such as rockfalls, and intrusion detection along railway tracks. A single optical fibre thus acts as a dense array of thousands of distributed sensors.

Challenge

DAS inherently provides one-dimensional localisation: it identifies the position of an event along the fibre, but not its distance from the fibre. To assess an alert correctly, it is essential to know whether the source is on the track, nearby or farther away, and on which side of the fibre it is located.

Internship objective

Design, compare and validate methods for two-dimensional localisation (position along the fibre and perpendicular distance) of vibration events in DAS data, including intrusions, rockfalls and passing trains.

Internship tasks

The work will focus on several complementary topics:

- **Array signal processing** (beamforming and MUSIC): treat the fibre as an array of aligned sensors to estimate source position both along and away from the fibre.
- **Deep learning**: train neural networks on time–distance representations to estimate the position (x, d).
- **Fibre geometry**: investigate how the layout (one or two fibres, straight or curved route) affects localisation accuracy and left/right ambiguity.
- **Method comparison**: measure accuracy, noise robustness, computational cost and real-time suitability as functions of signal-to-noise ratio and gauge length.

The internship will follow several stages:

- Review the literature on DAS source localisation;
- Become familiar with DAS data and the fibre installation geometry;
- Implement conventional methods followed by neural-network methods;
- Validate the approaches using controlled experiments and real-world data, then analyse performance.

Candidate profile

Education

Final-year engineering student, second-year Master's student or student on a gap year, with a background in signal processing, artificial intelligence or applied mathematics.

Technical skills

- Strong foundations in mathematics, signal processing and/or *machine learning*;
- Proficiency in Python (NumPy, SciPy and PyTorch are appreciated);
- Ability to analyse experimental data.

Personal qualities

- Scientific rigour, curiosity and independence;
- Teamwork and good communication skills.

Practical information

- Duration: 5 to 6 months
- Location: La Plaine-Saint-Denis, France
- Start date: February–March 2027
- Supervision: BEN AMARA Imen, RIHANI Mahran
- Allowance: in accordance with applicable regulations
- Benefits: SNCF travel benefits.

How to apply

Please send your CV and cover letter to all the following addresses:

- `ext.imen.benamara@reseau.sncf.fr`
- `gabriel.papaiz@reseau.sncf.fr`
- `ext.mahran.rihani@reseau.sncf.fr`

References

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- Liang J. et al. (2019). *Distributed acoustic sensing for 2D and 3D acoustic source localization.* Optics Letters, 44(7), 1690–1693.
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